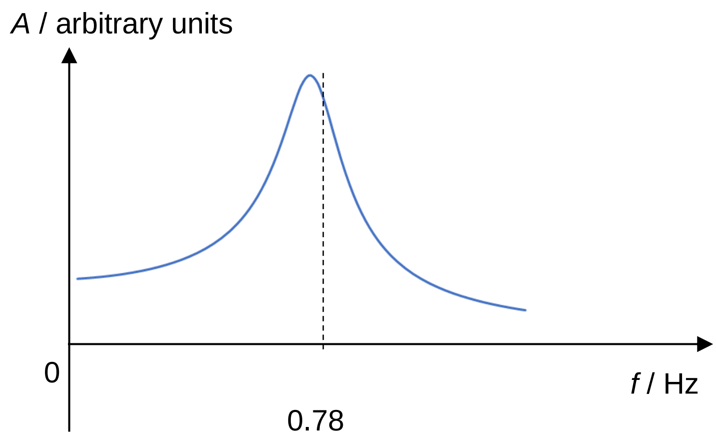
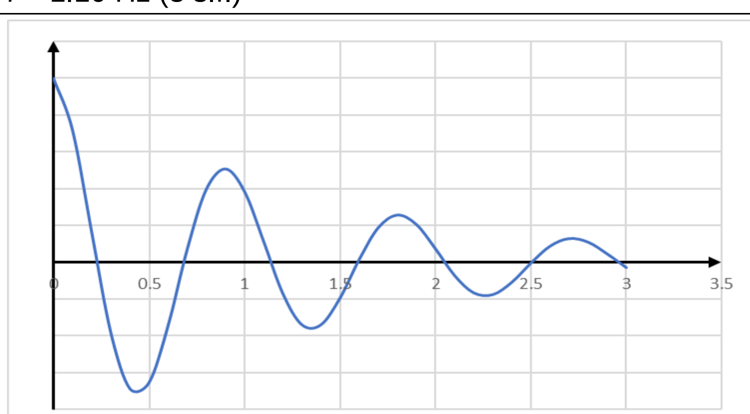
Q4	
(a)(i)	Taking the direction of displacement as positive, $F_R = mg - T$ $ma = mg - k(d + x)$
	Since $kd = mg$ $ma = -kx$ $a = -\dfrac{k}{m}x$
(a)(ii)	Negative sign of expression shows that acceleration acts in the opposite direction of the displacement.
	Since k and m are constants, acceleration is directly proportional to displacement.

(a)(iii)	$a = -\frac{k}{m}x$ $\omega^2 = \frac{k}{m}$ $(2\pi f)^2 = \frac{1.2}{50 \times 10^{-3}}$
	f = 0.780 Hz (3 s.f.)
(a)(iv)	 Correct shape, left side must be higher than right side, not symmetrical, cannot start from origin Peak slightly to the left of 0.78 Hz
(b)(i)	$k_2 = 2k$ $\omega^2 = \frac{2k}{m}$ $(2\pi f)^2 = \frac{2.4}{50 \times 10^{-3}}$
	f = 1.10 Hz (3 s.f.)
(b)(ii)	 Oscillating graph with period of about 0.91 s or slightly longer, must start from x_o at t = 0 Amplitude is decreasing exponentially with time
Q5	